

Navigators Dataset & Data Provenance

Benchmark Dataset Registries, Modalities, Licensing & Provenance Standards

Dataset Registry, Provenance, and Normalization

Registry Engine: `src/data/dataset_registry.py`
Normalization Engine: `src/data/sensor_normalization.py`

Dataset Classification and Provenance Standard

Navigators maintains strict separation between third-party external benchmark datasets and original Navigators datasets to guarantee zero provenance ambiguity.

Every dataset in the registry is registered with verifiable metadata:

- **dataset_name**: Official identifier.
- **source_url**: Verifiable publisher or repository URL.
- **license**: Open-source license specification.
- **citation**: Required academic citation.
- **country_code**: Geographic origin.
- **sampling_rate_hz**: Ingestion sampling rate.
- **ground_truth_type**: High-precision RTK GNSS, Optical Motion Capture, or Reference INS.
- **provenance_status**: VERIFIED or QUARANTINED.

1. External Benchmark Datasets

A. IO-VNBD (Vehicle Navigation Benchmark Dataset)

- **Source**: Smartphone In-Vehicle Dataset (Italy).
- **URL**: <https://github.com/m-pizzoli/IO-VNBD>
- **License**: Creative Commons Attribution 4.0 International (CC BY 4.0).
- **Modality**: 6-DOF IMU (accel, gyro), GNSS ground truth.
- **Sampling Rate**: 100 Hz IMU downsampled to 10-50 Hz.
- **Role**: Vehicle baseline model training and ablation studies.

B. RoNIN (Robust Network for Inertial Navigation)

- **Source**: Pedestrian Inertial Dataset (Simon Fraser University).
- **URL**: <https://github.com/ronin-rr/ronin>
- **License**: MIT License.
- **Modality**: 6-DOF IMU, 3D Magnetometer, 3D Pose GT via Vicon optical capture.
- **Sampling Rate**: 200 Hz downsampled to 100 Hz / 10 Hz.
- **Role**: Pedestrian baseline model training.

C. OxIOD (Oxford Inertial Odometry Dataset)

- **Source**: University of Oxford.
- **URL**: <https://oxiod.cs.ox.ac.uk/>
- **License**: Creative Commons Attribution 4.0 International.
- **Modality**: Handheld, pocket, trolley, and bag walking IMU sequences.

- **Sampling Rate:** 100 Hz.
- **Role:** Multi-placement pedestrian model training.

2. Original Navigators Datasets

A. Navigators India Vehicle Dataset

- **Geography:** Urban, rural, highway, and hilly terrain across India.
- **Vehicle Types:** Auto-rickshaws, compact hatchbacks, sedans, SUVs, motorcycles.
- **Modality:** 6-DOF IMU, 3D Magnetometer, Reference GNSS speed/heading.
- **Role:** Indian domain adaptation and generalization evaluation.

B. Navigators India Pedestrian Dataset

- **Geography:** Crowded markets, outdoor footpaths, indoor corridors in India.
- **Placements:** Handheld, front pocket, back pocket, shoulder bag.
- **Modality:** 6-DOF IMU, 3D Magnetometer, Reference RTK GNSS.
- **Role:** Pedestrian domain fine-tuning and benchmark evaluation.

3. Quarantined Datasets

- **I2WDD:** Quarantined. Although it contains Indian vehicle driving video/sensor data, its primary focus is driving behavior classification rather than continuous high-precision inertial odometry. It is preserved in registry quarantine without inclusion in default model training runs.

4. Cross-Dataset Normalization Rules

- **Zero Data Leakage:** Normalization statistics (mean, std, min, max) are calculated strictly on **TRAIN** splits and applied to validation and test splits without leaking future data.
- **Unit Conversions:** Standardized to SI units:
 - Accelerometer: m/s^2 (gravity constant $g = 9.80665 \text{ m/s}^2$).
 - Gyroscope: rad/s (converted from degrees per second where necessary).
- **Monotonic Timestamps:** Duplicate or out-of-order timestamps are strictly rejected during ingestion.